



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

A.Y. 2026 – 2027

B.Tech. III Year Odd Semester

COURSE: CNAD (24CND3101)

PROJECT ABSTRACT SUBMISSION

TITLE OF THE PROJECT	Cloud-Native Grocery Delivery System with Intelligent Budget Guardian	
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Abstract

Online grocery delivery systems provide convenience, but users often face difficulties in controlling their spending, tracking grocery expenses, and making cost-effective purchasing decisions. Traditional grocery platforms mainly concentrate on product browsing, cart management, ordering, payment, and delivery, while budget management is generally handled manually or through separate applications, providing limited real-time guidance. This project proposes a **Cloud-Native Grocery Delivery System with Intelligent Budget Guardian** to integrate grocery delivery with proactive and intelligent budget management. The system enables users to authenticate, browse and search products, add items to a cart, place and track orders, and make secure payments while setting a personal grocery budget. The Budget Guardian monitors cart spending in real time, provides alerts when users approach their budget limit, and offers smart suggestions and cheaper alternatives to support informed purchasing decisions. The proposed system follows a cloud-native microservices architecture using React, REST APIs, Docker, Kubernetes, NGINX Ingress, PostgreSQL, Redis, and Apache Kafka, supported by CI/CD using Jenkins and Docker and monitoring through Prometheus, Grafana, and ELK Stack. The expected outcome is a scalable, reliable, and intelligent grocery platform that improves user convenience while promoting responsible spending.

Signature of the Guide

Signature of the Course Coordinator

HOD

Tools and Technologies Used for the Project:

Category	Tools / Technologies
Frontend	HTML5, CSS3, JavaScript, React.js
Backend	Java Spring Boot
Database	MySQL / PostgreSQL
Version Control	Git, GitHub
Containerization	Docker
Container Orchestration	Kubernetes / Minikube
Package Management	Helm
CI/CD	GitHub Actions / Jenkins
Security Scanning	Trivy
Cloud Platform	AWS / Microsoft Azure / Google Cloud
API Testing	Postman
Development IDE	Visual Studio Code
Monitoring	Prometheus, Grafana
Documentation	GitHub / Markdown

Signature of the Guide**Course Coordinator****HOD**